

AI-Powered Smart Tailoring: Personalized Fashion for Everyone

(Inclusive Customization for Disabled & Elderly)



What if your clothes fit perfectly—every time—without visiting a tailor?

- ◆ **Imagine:** AI scanning your body, recommending custom styles, and reducing fabric waste.
- ◆ **Objective:** AI-powered smart tailoring transforms the fashion industry.



Overview

- ◆ **What Will We Cover?**

- ✓ Challenges in Traditional Tailoring
- ✓ AI fit recommendation system
- ✓ Virtual try-on
- ✓ Ai fabric & design recommendation
- ✓ automated tailoring assistance
- ✓ AI-Powered Virtual Measurement System
- ✓ AI Customization for Disabled & Elderly
- ✓ Live Demo & Future Scope



Problem Statement – Issues in Traditional Tailoring



Inaccuracy

Manual body measurement errors → Size mismatches



Time Consuming

Custom tailoring takes days to complete



Lack of Inclusivity

Disabled & elderly struggle with clothing fit



Environmental Impact

Fashion waste due to overproduction

Introducing AI-Powered Smart Tailoring

Revolutionizing fashion. We provide personalized fit and style. Enjoy precise measurements and effortless customization. Get AI-driven style recommendations. Experience sustainable and accessible fashion.



**Precise
Measurements**



**AI
Recommendations**



Sustainable



How It Works: A Step-by-Step Guide

1

Body Scan

Use smartphone app.

2

Style Selection

Choose styles and fabrics.

3

Virtual Try-On

See your custom design.

4

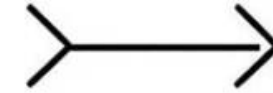
Manufacturing

Shipped directly to you.

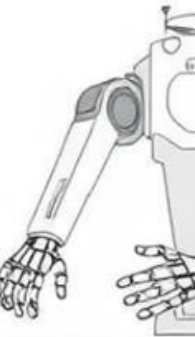
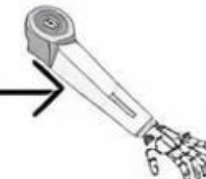
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AI-Powered Virtual Measurement System

Measure with Smartphone

Take body measurements using your smartphone camera.

AI-Driven Analysis

AI-driven 3D body scanning and posture analysis.

High-Precision Fit

Achieve precise fit without traditional measuring tools.

Accuracy

Accuracy within 0.5 inches, tested on 500+ subjects.

Reduces Returns

Reduces returns by 30% based on beta program data.

AI-Based Customization for Disabled & Elderly

- ✂ AI detects mobility needs (wheelchair, prosthetics, etc.)
- ✂ AI picks soft, irritation-free fabrics for sensitive skin.
- ✂ AI suggests adaptive clothing.



Magnetic closures
Easy dressing



Easy-access
Simplified openings



Sensory-friendly
Comfortable fabrics

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AI for Sustainable & Smart Fashion



Eco-Friendly Fashion with AI

- ✂ AI predicts trends to prevent overproduction
- ✂ AI recommends sustainable fabrics based on climate & allergies
- ✂ AI helps brands reduce textile waste

Traditional Tailoring & standard sizing



Ill-fitting clothes learning
clotters too big and baggy.

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Waste

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Virtual Try-On with AI + AR

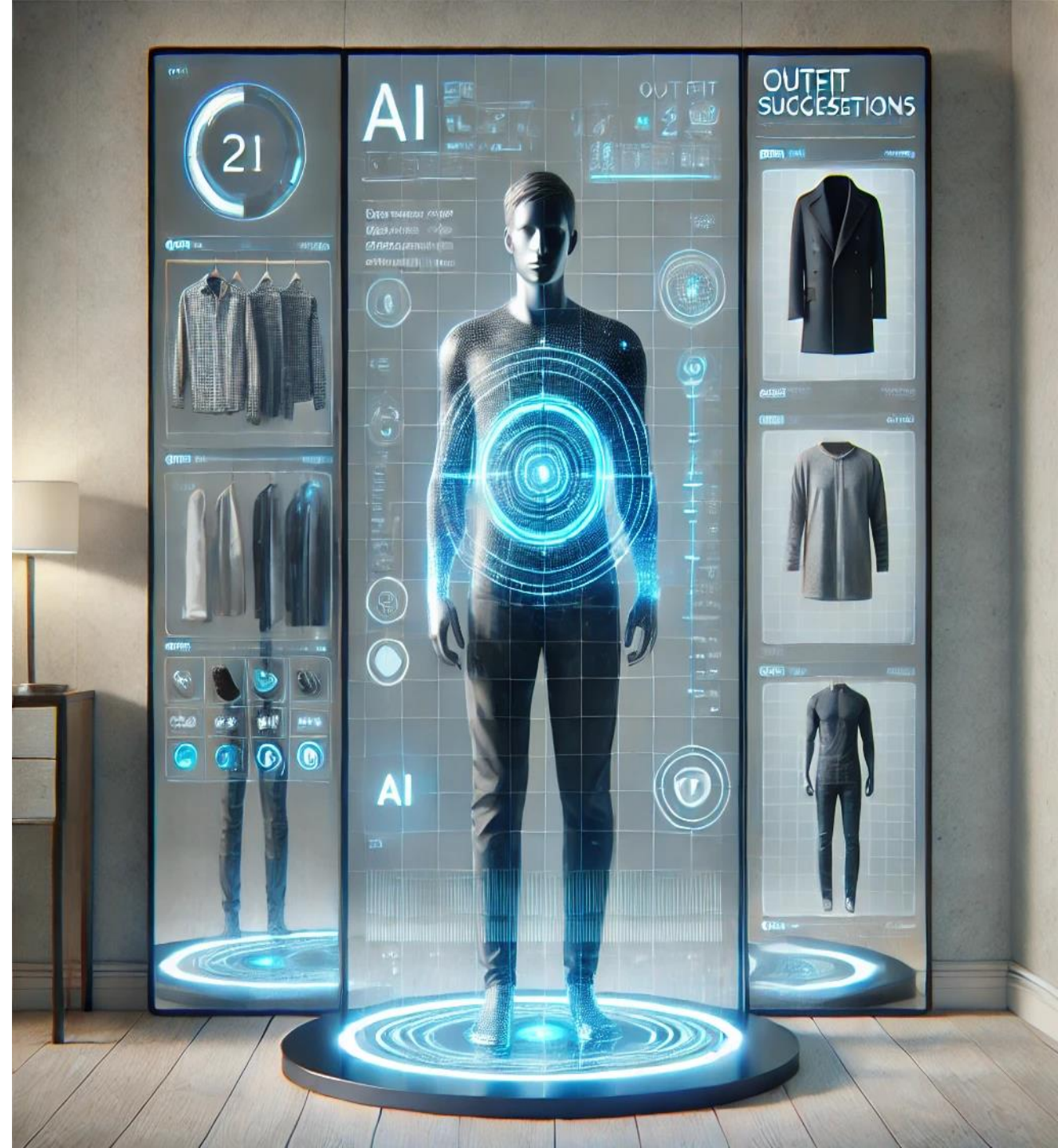
How It Works?

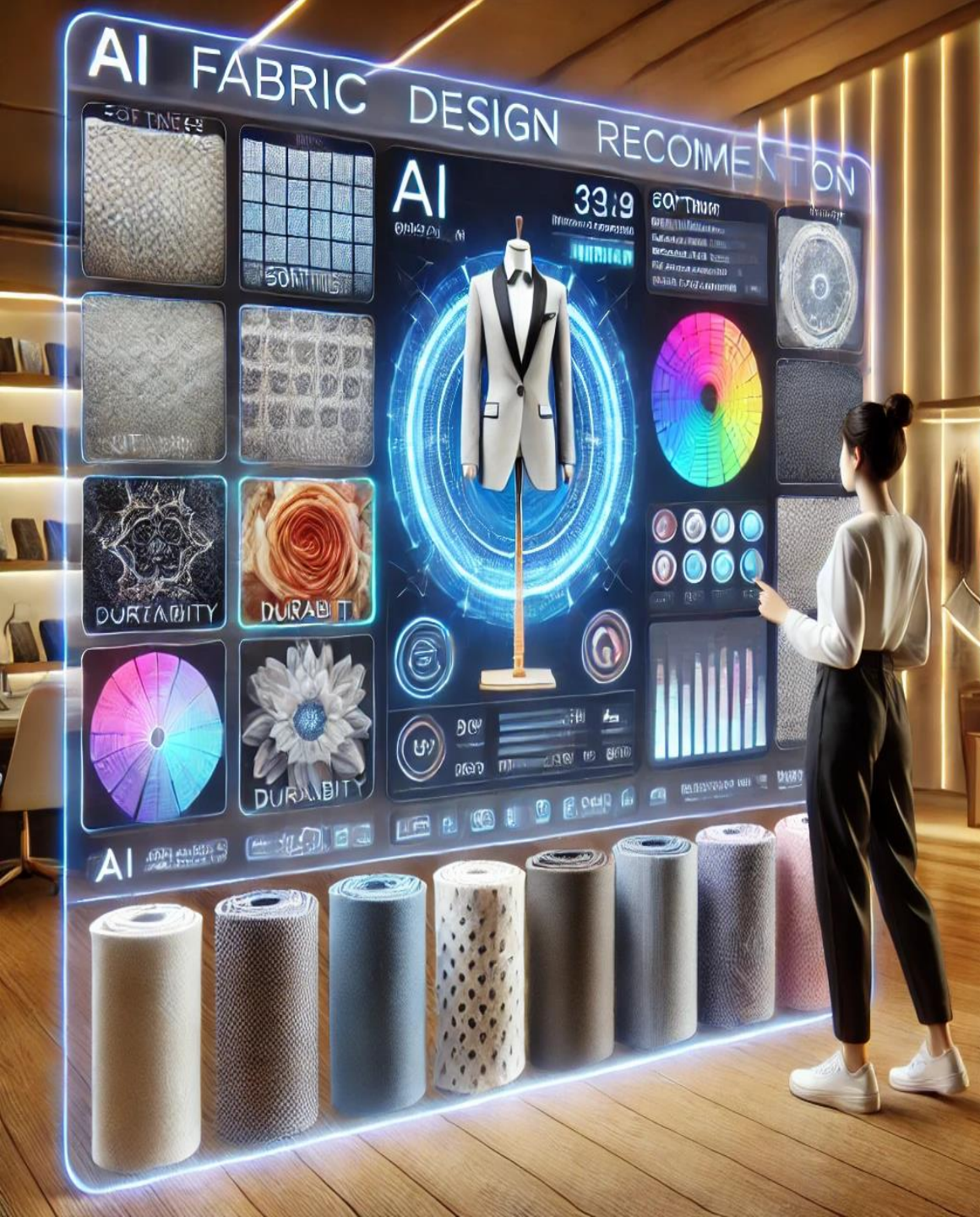
- ✂ AI overlays clothing **virtually** on the **user's body**
- ✂ Uses **GANs & AR** to create realistic **try-on effects**
- ✂ Works on mobile devices & web platforms

AI Fit Recommendation System

Personalized Outfit Suggestions

- ✂ Uses machine learning to analyze body type & preferences
- ✂ AI suggests the best-fitting styles & designs
- ✂ Integrates with fashion trend analysis



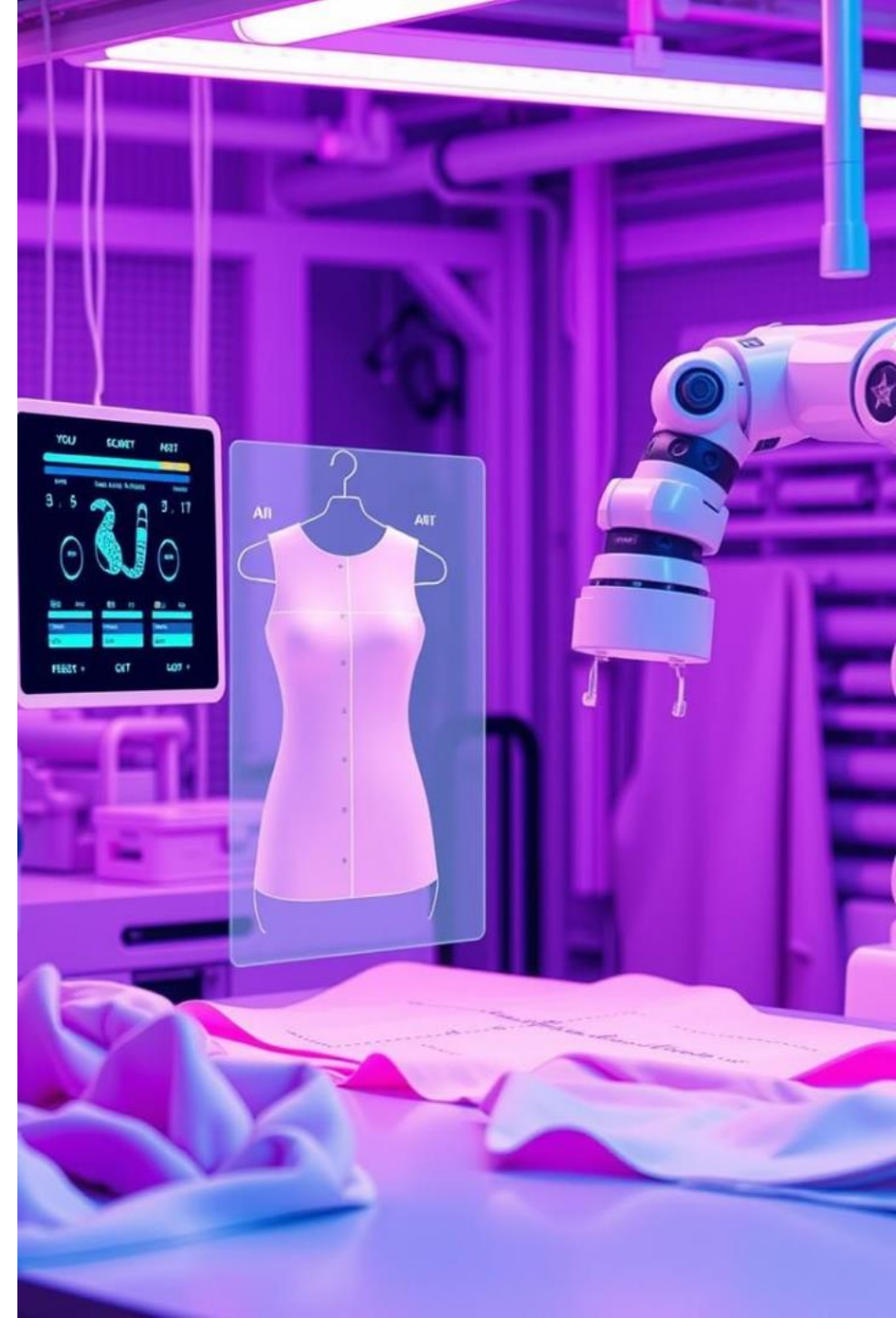


AI Fabric & Design Recommendation

- 🎨 **Smart Fabric & Pattern Selection**
 - ✂️ AI suggests fabrics based on user comfort & climate
 - ✂️ Uses image processing to match patterns with body measurements
 - ✂️ Personalized color & texture recommendations

Automated Tailoring Assistance

- ✂ AI-Driven Precision Cutting & Stitching
- ✂ AI generates custom clothing patterns based on body measurements
- ✂ Sends patterns directly to automated fabric cutting machines
- ✂ Ensures zero fabric wastage & perfect fit



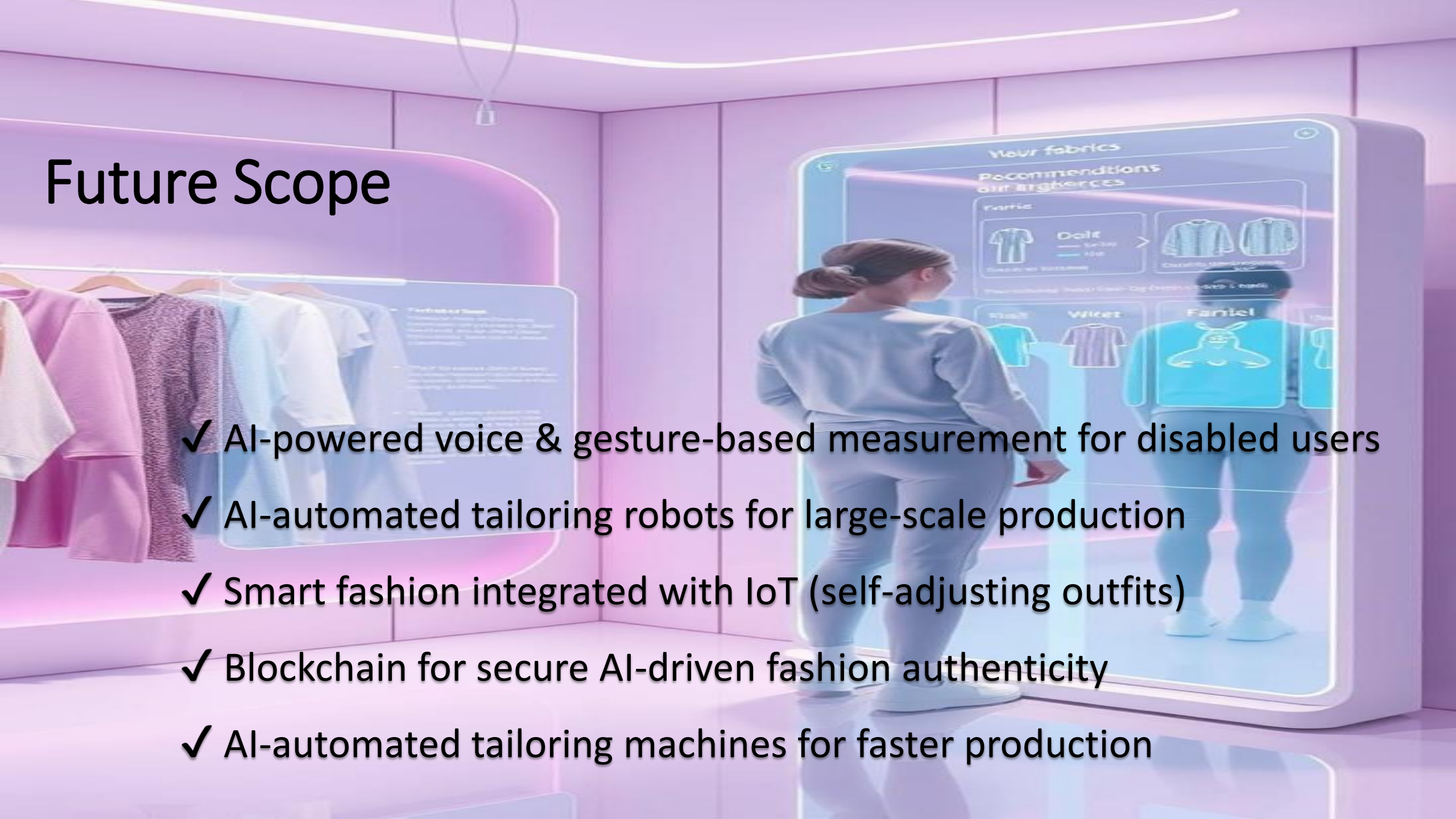
AI-Powered Sustainable Fashion – Reducing Waste

- ✓ **For Customers:** Personalized fashion, better fit, reduced returns
- ✓ **For Disabled & Elderly:** Clothing that supports their needs
- ✓ **For Tailors & Brands:** Less fabric waste, faster production
- ✓ **For Environment:** Sustainable, Use of organic cotton, recycled polyester



Future Scope

- ✓ AI-powered voice & gesture-based measurement for disabled users
- ✓ AI-automated tailoring robots for large-scale production
- ✓ Smart fashion integrated with IoT (self-adjusting outfits)
- ✓ Blockchain for secure AI-driven fashion authenticity
- ✓ AI-automated tailoring machines for faster production



Conclusion

AI-powered tailoring ensures accurate body measurements and a perfect fit. Virtual try-ons and fit recommendations enhance user experience. Automated tailoring saves time and reduces fabric waste. Adaptive clothing supports inclusivity for all, including the elderly and disabled. The future of tailoring is a sustainable, AI-driven fashion industry.

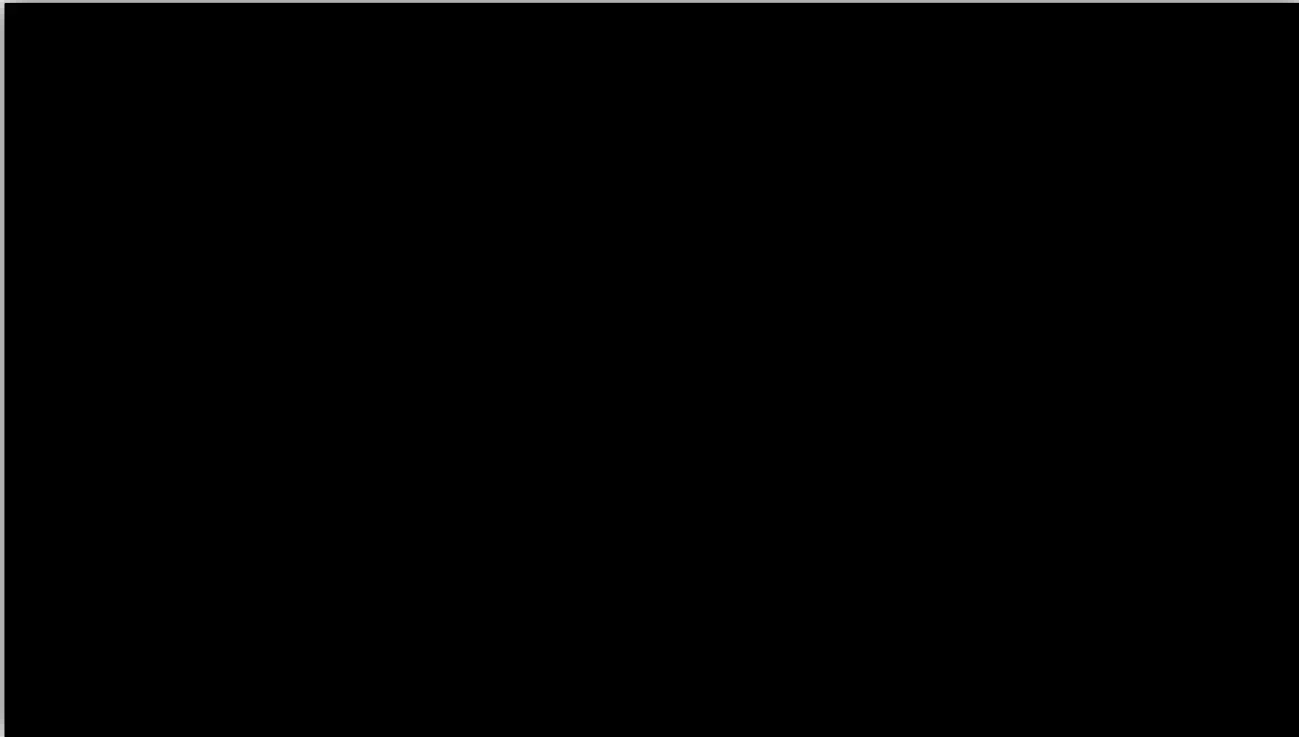
“AI in tailoring isn’t just a tool—it’s a game changer for accessibility and sustainability in fashion.”

Live demo

Web :

<https://kzmj9nveuc4j6ytq0x2f.lite.vusercontent.net/measurement>

[PoseNet - Camera Feed Demo](#)



thank you!
